Tutorial - 9 26/3/26

CROSS VALIDATION TECHNIQUE

1. Divide the dataset into 5folds for K-fold cross validation (4 samples per fold). list the customer ID's in each fold ensuring balanced class distribution.

2. Perform 5-fold cross validation for KNN with K=3 for fold 1 as validation set show
   - Training samples (16 customers)
   - validation samples (4 customers)
   - calculate Education distances for at least one validation sample
   - predict class for all validation samples calculate fold 1 accuracy.

3. complete the c-v for all 5 folds and create a summary table

4. calculate the overall cross validation metrics
   - Mean cv Accuracy
   - standard Deviation of Accuracy
   - 95% confidence Interval for accuracy

customerID	Age	Annual Income	Shopping frequency	Puchase
1	25	50	8	Y
2	45	80	3	N
3	28	55	9	Y
4	50	85	2	N
5	30	60	10	Y
6	48	82	3	N
7	26	52	8	Y
8	47	88	2	N
9	29	58	9	Y
10	46	84	3	N
11	27	54	8	Y
12	49	86	2	N
~~13~~				
~~14~~				
~~15~~				
16				
17				
18				
19				
20				

$\frac{20}{5} = 4$ 20 → 5 fold 14 - fold

(16, 4) CV

fold - 1

12 → data 4 → folds

$\frac{12}{4} = 3$ (9, 3)

Training Validation

KNN (K=5)

1) Distance for (46, 84, 3)

(46, 84, 3)
- (25, 50, 8)
- (45, 80, 3)
- (28, 55, 4)
- (50, 85, 2)
- (30, 60, 10)
- (48, 82, 3)
- (26, 52, 8)
- (47, 88, 2)
- (29, 58, 4)

$$\text{distance} = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

$$(46, 84, 3)(25, 50, 8) = \sqrt{(46, 25)^2 + (84 - 50)^2 + (3 - 8)^2}$$

40.28
4.12 ✓ N
34.60
4.24 ✓ N 5 - Nearest
29.83 ✓ Y
2.83 ✓ N
37.11
4.24 ✓ N
32.45

∴ N = 4
Y = 1

prediction = N

2) Distance for (27, 54, 8)

4.47 ✓ Y
32.02
1.73 ✓ Y
39.08
7.00 ✓ Y
25.36
2.24 ✓ Y
39.87
4.58 ✓ Y

Y = 5
N = 0

Prediction = Y

3) Distance for (49, 56, 2)

93.68

7.28 ✓ N

38.10

1.41 ✓ N

33.18 ✓ Y

4.24 ✓ N

41.49

2.23 ✓ N

35.11

N = 4

Y = 1

Prediction = N

Validation point	Actual	predicted
(46, 84, 3)	N	N
(27, 54, 8)	Y	Y
(49, 86, 2)	N	N

∴ ~~PRE~~ Accuracy = 100%